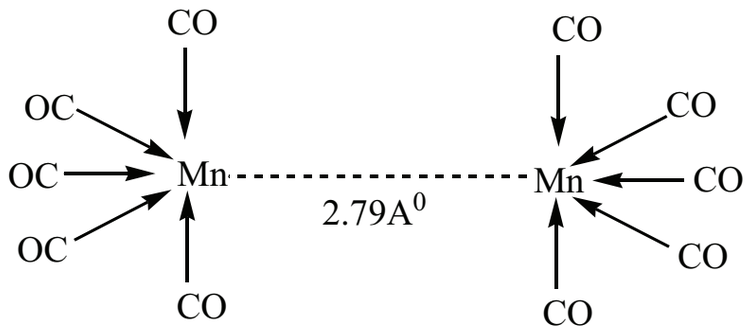
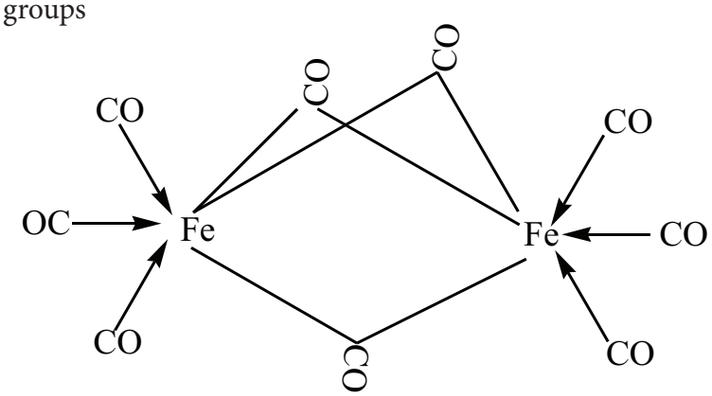


Other examples of this type are, $Tc_2(CO)_{10}$, and $Re_2(CO)_{10}$.

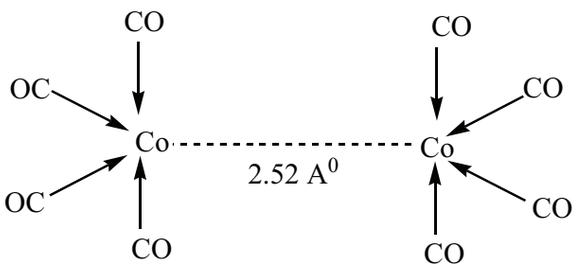
b. Bridged carbonyls:

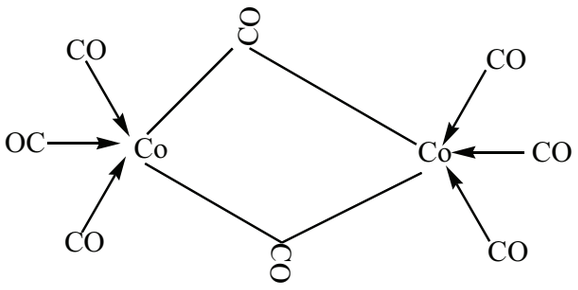
These metal carbonyls contain one or more bridging carbonyl ligands along with terminal carbonyl ligands and one or more Metal-Metal bonds. For example,

(i) The structure of $Fe_2(CO)_9$, di-iron nona carbonyl molecule consists of three bridging CO ligands, six terminal CO groups



(ii) For dicobaltoctacarbonyl $Co_2(CO)_8$ two isomers are possible. The one has a metal–metal bond between the cobalt atoms, and the other has two bridging CO ligands.





Bonding in metal carbonyls

In metal carbonyls, the bond between metal atom and the carbonyl ligand consists of two components. The first component is an electron pair donation from the carbon atom of carbonyl ligand into a vacant d-orbital of central metal atom. This electron pair donation forms $M \xleftarrow{\sigma \text{ bond}} CO$ sigma bond. This sigma bond formation increases the electron density in metal d orbitals and makes the metal electron rich. In order to compensate for this increased electron density, a filled metal d-orbital interacts with the empty π^* orbital on the carbonyl ligand and transfers the added electron density back to the ligand. This second component

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